

Non-Functional Requirements

1. Performance

- Pages should load within approximately 3 seconds under normal conditions.
- The system should handle multiple users while maintaining stable performance.
- Loading indicators should be displayed during long-running actions.

2. Security

- Passwords should be securely hashed and stored.
- All communication should use HTTPS.
- User data should be protected from unauthorized access.
- API keys and sensitive credentials should be stored securely.

3. Usability

- The system should provide a simple and user-friendly interface.
- The interface should be responsive across desktop, tablet, and mobile devices.
- Navigation should be intuitive and easy to use.
- The system should provide clear feedback for actions such as saving, errors, and loading.

4. Reliability and Availability

- The system should aim for high availability with minimal downtime.
- The system should handle errors gracefully without crashing.
- The system should preserve user data during navigation or unexpected interruptions.

5. Compatibility

- The system should work on modern browsers such as Chrome, Firefox, Edge, and Safari.
- The system should support different screen sizes and resolutions.

6. Maintainability

- The system should be modular and easy to update.
- Code should be structured for easy debugging and maintenance.

- Components and styles should be reusable.

7. API and Integration

- The system should handle GitHub and AI API failures gracefully.
- The system should manage API rate limits efficiently.
- The system should validate and sanitize API responses before displaying data.

8. Accessibility

- The system should use readable typography and sufficient color contrast.
- Interactive elements should be accessible through keyboard navigation.
- Forms should include labels, placeholders, and validation messages.
- The interface should follow accessibility best practices for all users.